

Nour Bassem

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EDUCATION

Misr International University

Bachelor's Degree in Computer Science, Minor in Artificial Intelligence; GPA: 3.4/4.0

Cairo, Egypt

Sep. 2022 – Jun. 2026

EXPERIENCE

Deloitte

Computer Science Intern

- Rotated across Artificial Intelligence, Full-Stack Development, Cybersecurity, Cloud Systems, and Software Testing, gaining exposure to enterprise technology workflows.
- Built [FinanceHub](#), an LLM-powered financial analysis tool using prompt engineering and RAG to generate structured insights and support financial reporting.
- Presented the tool as a capstone project and collaborated with cross-functional teams in an agile environment.

Cairo, Egypt

Jul. 2025 – Sep. 2025

International Exchange Student Internship – UTU Campus

Full-Stack Intern

- Developed responsive React.js and Next.js user interfaces using HTML, CSS, and JavaScript, while collaborating with designers and backend engineers to deliver scalable web applications.
- Strengthened cross-cultural communication, adaptability, teamwork, and problem-solving skills through an international exchange experience and daily interaction with students, faculty, and professionals in India.

Surat, India

Jul. 2023 – Sep. 2023

Misr International University

Co-Assistant

- Assisted and mentored in programming labs in debugging, algorithms, problem-solving, and software development concepts.

Cairo, Egypt

Oct. 2024 – Jan. 2025

PROJECTS

ReVerie – AI-Powered Therapeutic Platform for Alzheimer's Patients

Graduation Project; Supabase, Computer Vision, 3D Reconstruction, Unreal Engine, LaMa, WorldGen, GCP, Flutter/Dart

- Co-developed a digital therapeutic platform that transforms personal photographs into immersive VR reminiscence scenes using depth estimation, segmentation, inpainting, and 3D human/object/environment reconstruction.
- Worked on environment reconstruction and scene assembly by integrating generated environments, reconstructed humans/objects, backend APIs, and Unreal Engine visualization for supervised patient-doctor workflows.

Smart Bin Waste Management System

Intelligent agents; TensorFlow/Keras, DenseNet121, OpenCV, A* Search, Arduino, Ultrasonic Sensors, Servo/Stepper Motors

- Built a two-agent waste-management prototype combining computer-vision waste classification, fill-level monitoring, overflow prediction, and route optimization.
- Fine-tuned DenseNet121 on a 2,527-image, 4-class garbage dataset, achieving 98% test accuracy and 97.89% macro F1; integrated A* search for pickup-route planning.

Multi-Label Emotion Intelligence System

NLP; PyTorch, Hugging Face, RoBERTa/BERT, Multi-Label Classification, Threshold Tuning

- Fine-tuned a transformer-based multi-label emotion classifier using class-weighted BCE loss, per-label threshold tuning, and sarcasm-aware interpretation.
- Achieved 0.598 micro-F1, 0.964 hamming accuracy, and 0.555 sample Jaccard on GoEmotions; trained a separate sarcasm classifier with 0.635 F1.

Food Calorie Estimation System

Deep learning; ResNet18, Pandas, NumPy, Image Segmentation, Pixel-Based Calibration, MiDaS Depth Estimation

- Developed an image-based calorie estimator using ResNet18 transfer learning for food classification and pixel-based calibration to estimate portion weight.
- Mapped predicted food labels to nutritional values and computed calories from estimated grams; tested MiDaS depth estimation as a future scaling option.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, C++, Java, Dart, HTML/CSS, PHP

AI/ML/Data: Machine Learning, Deep Learning, Computer Vision, NLP, RAG, LLMs, Prompt Engineering, Feature Engineering, Data Analysis, Generative AI, LangChain

Frameworks/Libraries: PyTorch, TensorFlow/Keras, Scikit-learn, Hugging Face, Pandas, NumPy, OpenCV, Flutter, Node.js, Express.js

Tools/Cloud: Git, GitHub, Supabase, MongoDB, MySQL, Google Colab, VS Code, Unreal Engine, Blender, GCP

RESEARCH

Emotion Detection Using NLP – IEEE Published Research

- Co-authored IEEE-published research on text-based emotion detection using machine learning and deep learning models.

AI-Driven Digital Therapeutic Interventions for Early-Stage Alzheimer's Disease | Submitted to *Complexity* (Wiley), 2025

- systematic review of AI-driven memory reconstruction, VR reminiscence therapy, computer vision, digital twins, and cognitive stimulation technologies for Alzheimer's disease.

ReVerie: Personalized AI-Based 2D-to-3D Memory Reconstruction for Immersive Reminiscence Therapy

- Submitted to MIUCC'26; proposed a personalized AI framework for transforming patient photos into immersive VR experiences.

LANGUAGES

Arabic: Native | English: C2 | German: B1 Goethe Certificate / B2 proficiency